

Research Findings Summary – NYC Airbnb Dataset

1. Geographic Patterns:

- Hexbin plots reveal dense clustering of Airbnb listings in Manhattan and Brooklyn.
- Prices tend to be higher in central Manhattan; spatial patterns show price hotspots.

2. Price Distribution:

- Price distribution is heavily right-skewed, indicating many low-priced listings and a few extremely expensive ones.
- Outlier removal significantly improved the clarity of visualizations.

3. Relationship Insights:

- Minimum nights vs. price showed no strong linear relationship but showed patterns after removing outliers.
- Reviews per month vs. price also displayed nonlinear relationships visible in hexbin density plots.

4. Feature Importance (SHAP Analysis):

- SHAP-based analysis highlighted the most influential factors affecting price prediction.
- Top drivers included:
 - Room type encoding variables
 - Neighbourhood group indicators
 - Geographic coordinates (latitude and longitude)
 - Minimum nights and min_price_pay
- Categorical encodings (room type, neighbourhood group) had strong predictive influence.

5. Model Behavior:

- XGBoost regressor successfully captured nonlinear relationships.
- SHAP visualizations (beeswarm and bar plots) showed that several features contribute positively or negatively depending on listing characteristics.

6. Data Cleaning Impact:

- Removing extreme outliers stabilized model behavior.
- Filling missing values improved dataset consistency for modeling.

Overall Conclusion:

The research shows that Airbnb prices in NYC are strongly influenced by location, room type, and minimum stay requirements. SHAP analysis provides interpretable insights, confirming that spatial and categorical factors dominate pricing dynamics.